S-I. SUPPLEMENTARY SECTION

A. Kinetic model expressions

Phosphotransferase system (PTS)

$$v_1 = \frac{v_{PTS}^{\max}\,[glc]\,\dfrac{[pep]}{[pyr]}}{K_{a1} + K_{a2}\dfrac{[pep]}{[pyr]} + K_{a3}[glc] + [glc]\,\dfrac{[pep]}{[pyr]}} \cdot \frac{1}{1 + \left(\dfrac{[g6p]^{n_{g6p}}}{K_{g6p}}\right)} \tag{20}$$

Empirical formulation from Kadir et al. [12]; thermodynamic driving force not applicable.

Glucose 6-phosphate isomerase (PGI) EC: 5.3.1.9 $g6p \rightleftharpoons f6p$

$$\kappa_2 = \frac{[g6p]/K_{s,g6p}^2}{1 + [g6p]/K_{s,g6p}^2 + [f6p]/K_{p,f6p}^2} \qquad \gamma_2 = 1 - \frac{[f6p]}{[g6p]\,K'_{eq,2}} \tag{21}$$

$$v_2 = Pgi \cdot k_{cat,2}^{+} \cdot \kappa_2 \cdot \gamma_2 \tag{22}$$

Phosphofructokinase (PFK) EC: 2.7.1.11 $f6p + atp \rightleftharpoons fbp + adp$

$$\kappa_3 = \frac{([f6p]/K_{s,f6p}^3)\,([atp]/K_{s,atp}^3)}{1 + ([f6p]/K_{s,f6p}^3)([atp]/K_{s,atp}^3) + ([fbp]/K_{p,fbp}^3)([adp]/K_{p,adp}^3)} \qquad \gamma_3 = 1 - \frac{[fbp]\,[adp]}{[f6p]\,[atp]\,K'_{eq,3}} \tag{23}$$

$$v_3 = PfkB \cdot k_{cat,3}^{+} \cdot \kappa_3 \cdot \gamma_3 \tag{24}$$

Fructose-bisphosphate aldolase (FBA) EC: 4.1.2.13 $fbp \rightleftharpoons g3p + dhap$

$$\kappa_4 = \frac{[fbp]/K_{s,fbp}^4}{1 + [fbp]/K_{s,fbp}^4 + ([g3p]/K_{p,g3p}^4)([dhap]/K_{p,dhap}^4)} \qquad \gamma_4 = 1 - \frac{[g3p]\,[dhap]}{[fbp]\,K'_{eq,4}} \tag{25}$$

$$v_4 = FbaA \cdot k_{cat,4}^{+} \cdot \kappa_4 \cdot \gamma_4 \tag{26}$$

Triosephosphate isomerase (TPI) EC: 5.3.1.1 $dhap \rightleftharpoons g3p$

$$\kappa_5 = \frac{[dhap]/K_{s,dhap}^5}{1 + [dhap]/K_{s,dhap}^5 + [g3p]/K_{p,g3p}^5} \qquad \gamma_5 = 1 - \frac{[g3p]}{[dhap]\,K'_{eq,5}} \tag{27}$$

$$v_5 = TpiA \cdot k_{cat,5}^{+} \cdot \kappa_5 \cdot \gamma_5 \tag{28}$$

Glyceraldehyde 3-phosphate dehydrogenase (GAPDH) EC: 1.2.1.12 $g3p + pi + nad \rightleftharpoons pgp + nadh$

$$\kappa_6 = \frac{([g3p]/K_{s,g3p}^6)([pi]/K_{s,pi}^6)([nad]/K_{s,nad}^6)}{1 + ([g3p]/K_{s,g3p}^6)([pi]/K_{s,pi}^6)([nad]/K_{s,nad}^6) + ([pgp]/K_{p,pgp}^6)([nadh]/K_{p,nadh}^6)} \tag{29}$$

$$\gamma_6 = 1 - \frac{[pgp]\,[nadh]}{[g3p]\,[pi]\,[nad]\,K'_{eq,6}} \tag{30}$$

$$v_6 = GapA \cdot k_{cat,6}^{+} \cdot \kappa_6 \cdot \gamma_6 \tag{31}$$

Phosphoglycerate kinase (PGK) EC: 2.7.2.3 $pgp + adp \rightleftharpoons 3pg + atp$

$$\kappa_7 = \frac{([pgp]/K_{s,pgp}^7)([adp]/K_{s,adp}^7)}{1 + ([pgp]/K_{s,pgp}^7)([adp]/K_{s,adp}^7) + ([3pg]/K_{p,3pg}^7)([atp]/K_{p,atp}^7)} \qquad \gamma_7 = 1 - \frac{[3pg]\,[atp]}{[pgp]\,[adp]\,K'_{eq,7}} \tag{32}$$

$$v_7 = Pgk \cdot k_{cat,7}^{+} \cdot \kappa_7 \cdot \gamma_7 \tag{33}$$

Phosphoglycerate mutase (GPM) EC: 5.4.2.11 $3pg \rightleftharpoons 2pg$

$$\kappa_8 = \frac{[3pg]/K_{s,3pg}^8}{1 + [3pg]/K_{s,3pg}^8 + [2pg]/K_{p,2pg}^8} \qquad \gamma_8 = 1 - \frac{[2pg]}{[3pg]\,K'_{eq,8}} \tag{34}$$

$$v_8 = GpmA \cdot k_{cat,8}^{+} \cdot \kappa_8 \cdot \gamma_8 \tag{35}$$

Enolase (ENO) EC: 4.2.1.11 $2pg \rightleftharpoons pep$

$$\kappa_9 = \frac{[2pg]/K^9_{s,2pg}}{1 + [2pg]/K^9_{s,2pg} + [pep]/K^9_{p,pep}} \qquad \gamma_9 = 1 - \frac{[pep]}{[2pg]\, K'_{eq,9}} \tag{36}$$

$$v_9 = Eno \cdot k^+_{cat,9} \cdot \kappa_9 \cdot \gamma_9 \tag{37}$$

The stoichiometric matrix of this system is:

$$\mathbf{S} = \begin{array}{c} \\ \text{2pg} \\ \text{3pg} \\ \text{dhap} \\ \text{f6p} \\ \text{fbp} \\ \text{g3p} \\ \text{g6p} \\ \text{pep} \\ \text{pgp} \end{array} \begin{array}{c} \begin{array}{ccccccccc} \text{pts} & \text{pgi} & \text{pfk} & \text{fba} & \text{tpi} & \text{gap} & \text{pgk} & \text{gpm} & \text{eno} \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & -1 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 \end{bmatrix} \end{array} \tag{38}$$

S-II. FIGURES

A. *Escherichia coli* metabolism

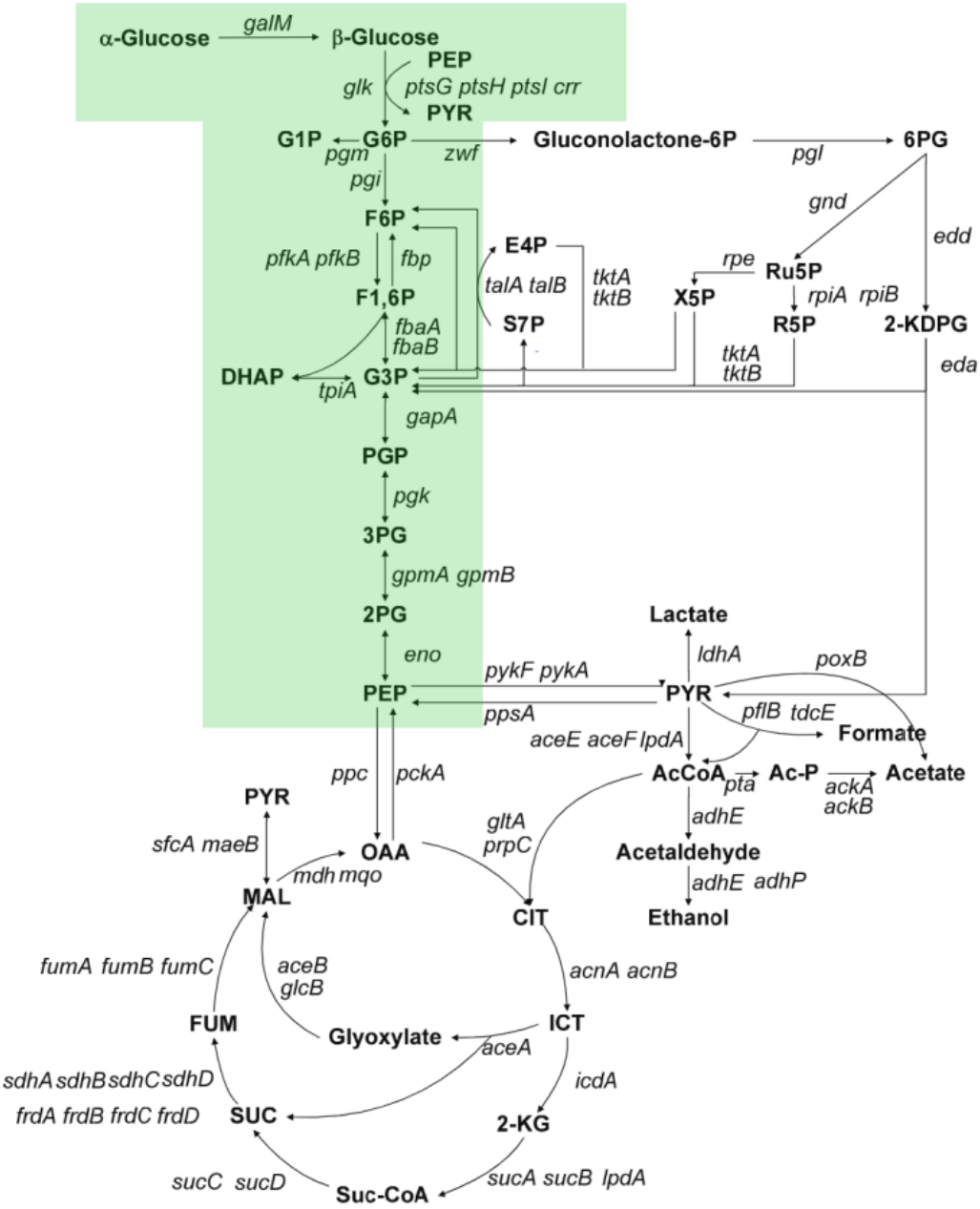


Figure S1: Carbon metabolism pathways of *E.coli* [2]. The reaction highlighted in green encompass the studied reactions.

B. Model descriptions

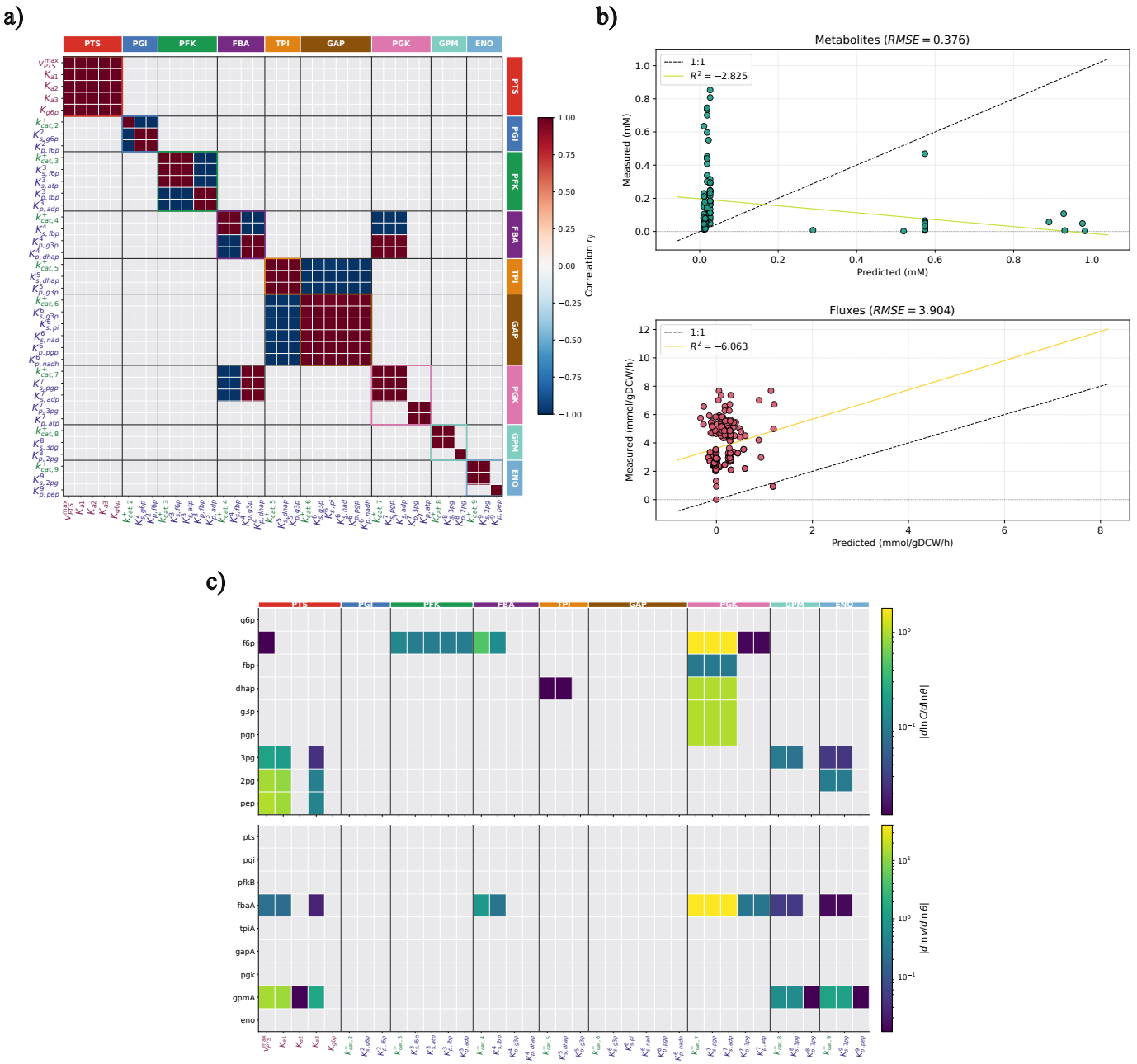


Figure S2: **Fixed-parameters (literature model) analysis**. **a)** FIM-derived parameter correlation matrix computed from the literature-fixed parameter set, grouped by reaction. **b)** Parity plots comparing measured vs. predicted metabolite concentrations (top, RMSE $= 0.376$, $R^2 = -2.825$) and fluxes (bottom, RMSE $= 3.904$, $R^2 = -6.063$) using literature parameter values without any data-driven fitting. **c** Relative sensitivity matrix of the model output to the 37 parameters of the model evaluated at the parameter value and model outputs.

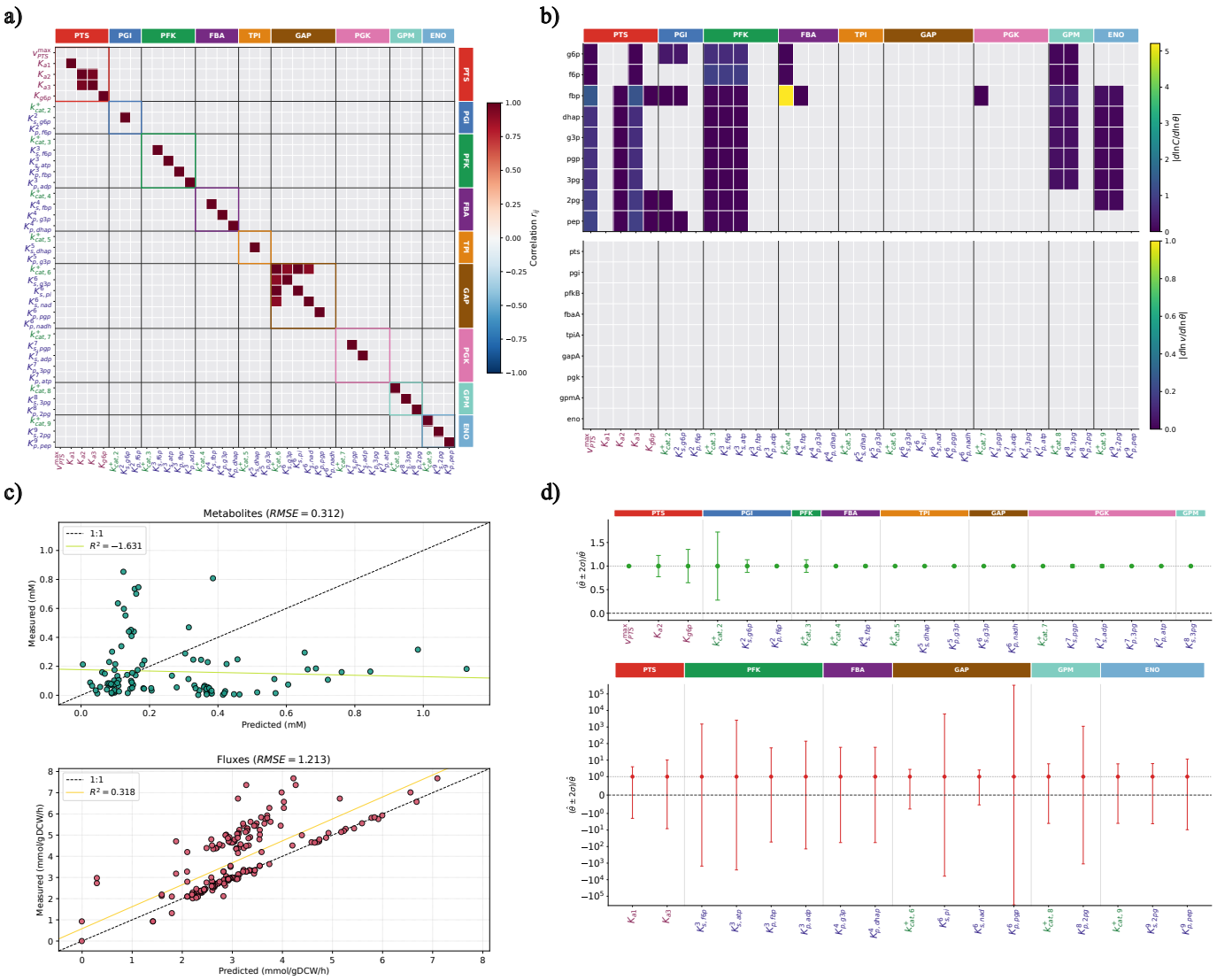


Figure S3: All-parameters (initial free) model. **a)** FIM-derived parameter correlation matrix for the 37 freely estimated parameters. **b)** Normalized sensitivity coefficients of metabolite concentrations (top, $|\partial \ln C/\partial \ln \theta|$) and fluxes (bottom, $|\partial \ln v/\partial \ln \theta|$) with respect to each parameter, grouped by reaction. **c)** Parity plots of measured vs. predicted metabolite concentrations (top, RMSE $= 0.312$) and fluxes (bottom, RMSE $= 1.213$) for the fitted 37-parameter model. **d)** FIM-derived relative confidence intervals, $(\hat{\theta} \pm 2\sigma)/\hat{\theta}$, per parameter; parameters with narrow, well-constrained intervals are shown in green (top), and those with wide or unbounded intervals in red (bottom, log scale).

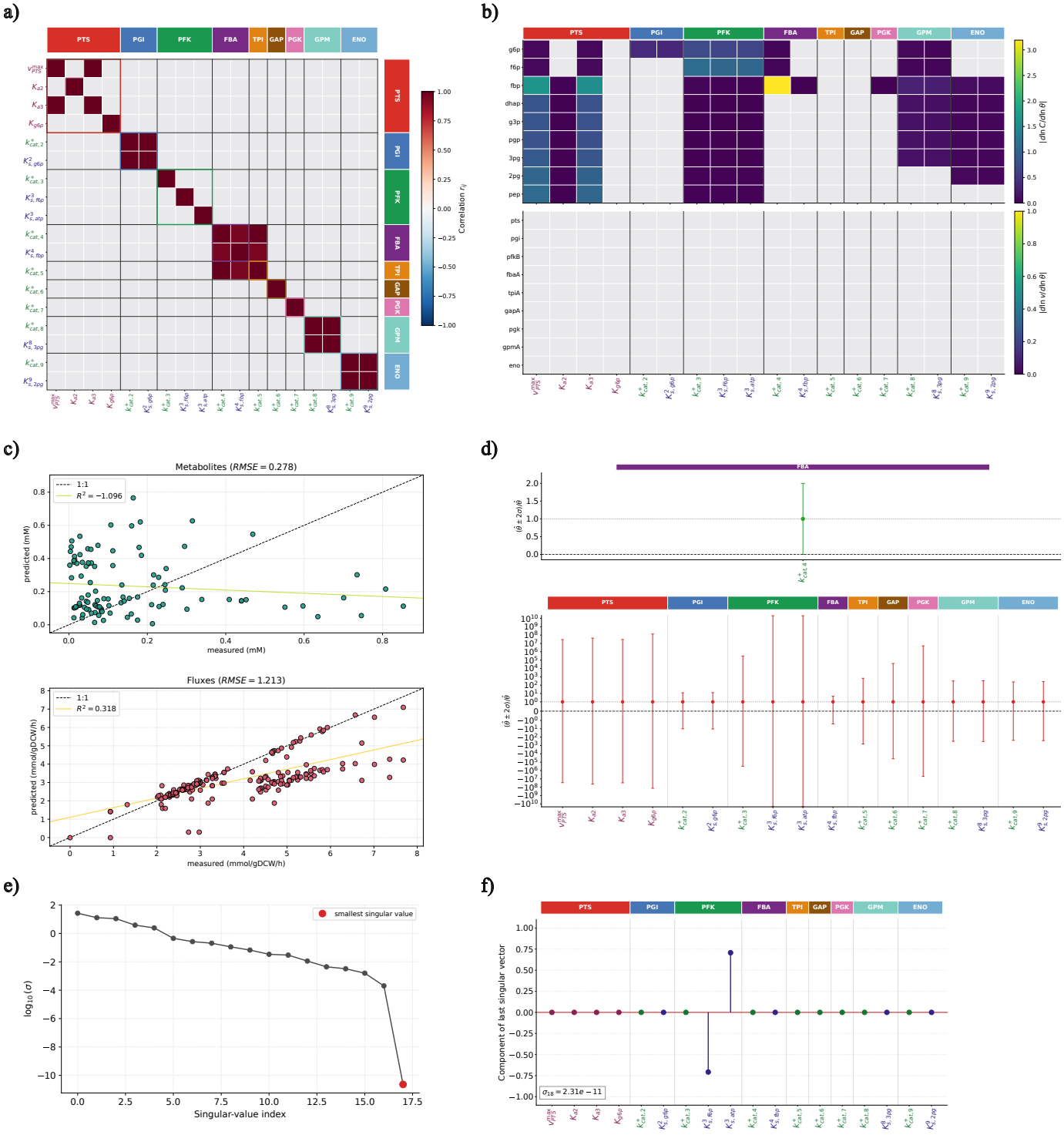


Figure S4: First reduced model (18 free parameters). **a)** FIM-derived correlation matrix after fixing 19 highly correlated, low-sensitivity parameters from the 37-parameter model (boxed blocks indicate parameters grouped by reaction). **b)** Normalized sensitivity coefficients of metabolites (top) and fluxes (bottom) for the 18 remaining free parameters. **c)** Parity plots of measured vs. predicted metabolite concentrations (top, RMSE $= 0.278$, $R^2 = -1.096$) and fluxes (bottom, RMSE $= 1.213$, $R^2 = 0.318$). **d)** FIM-derived relative confidence intervals, highlighting $k^+_{cat,4}$ (FBA) as a well-constrained parameter (top) against the full 18-parameter set (bottom, log scale). **e)** Singular value spectrum of the relative sensitivity matrix; the smallest singular value (red) reveals a near-null direction in parameter space. **f)** Contribution of each parameter to the singular vector associated with the smallest singular value, identifying $K^3_{s,f6p}$ and $K^3_{s,atp}$ (PFK) as the structurally redundant pair targeted in the second reduction step.

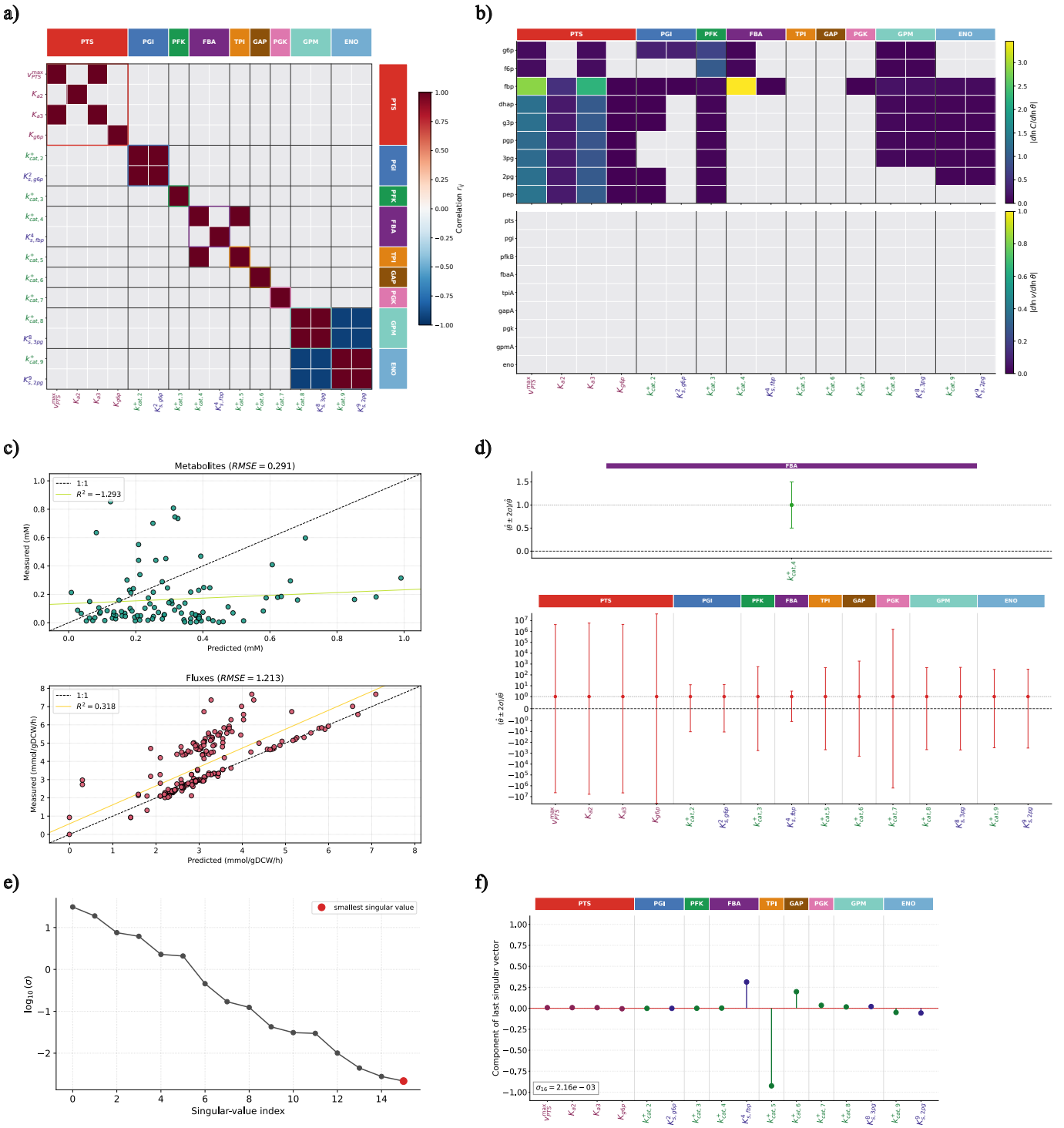


Figure S5: Final reduced model (16 free parameters). **a)** FIM-derived correlation matrix after the second reduction step. **b)** Normalized sensitivity coefficients of metabolites (top) and fluxes (bottom) for the 16 free parameters. **c)** Parity plots of measured vs. predicted metabolite concentrations (top, RMSE $= 0.291$, $R^2 = -1.293$) and fluxes (bottom, RMSE $= 1.213$, $R^2 = 0.318$). **d)** FIM-derived relative confidence intervals, highlighting $k^+_{cat,4}$ (FBA) as the most reliably constrained parameter (top) against the full parameter set (bottom, log scale). **e)** Singular value spectrum of the relative sensitivity matrix, confirming no further near-null directions beyond numerical noise ($\sigma_{16} = 2.16 \times 10^{-3}$). **f)** Contribution of each parameter to the singular vector associated with the smallest singular value, dominated by FBA and TPI parameters, consistent with the absence of additional structural redundancy.

C. Reduced model evaluation

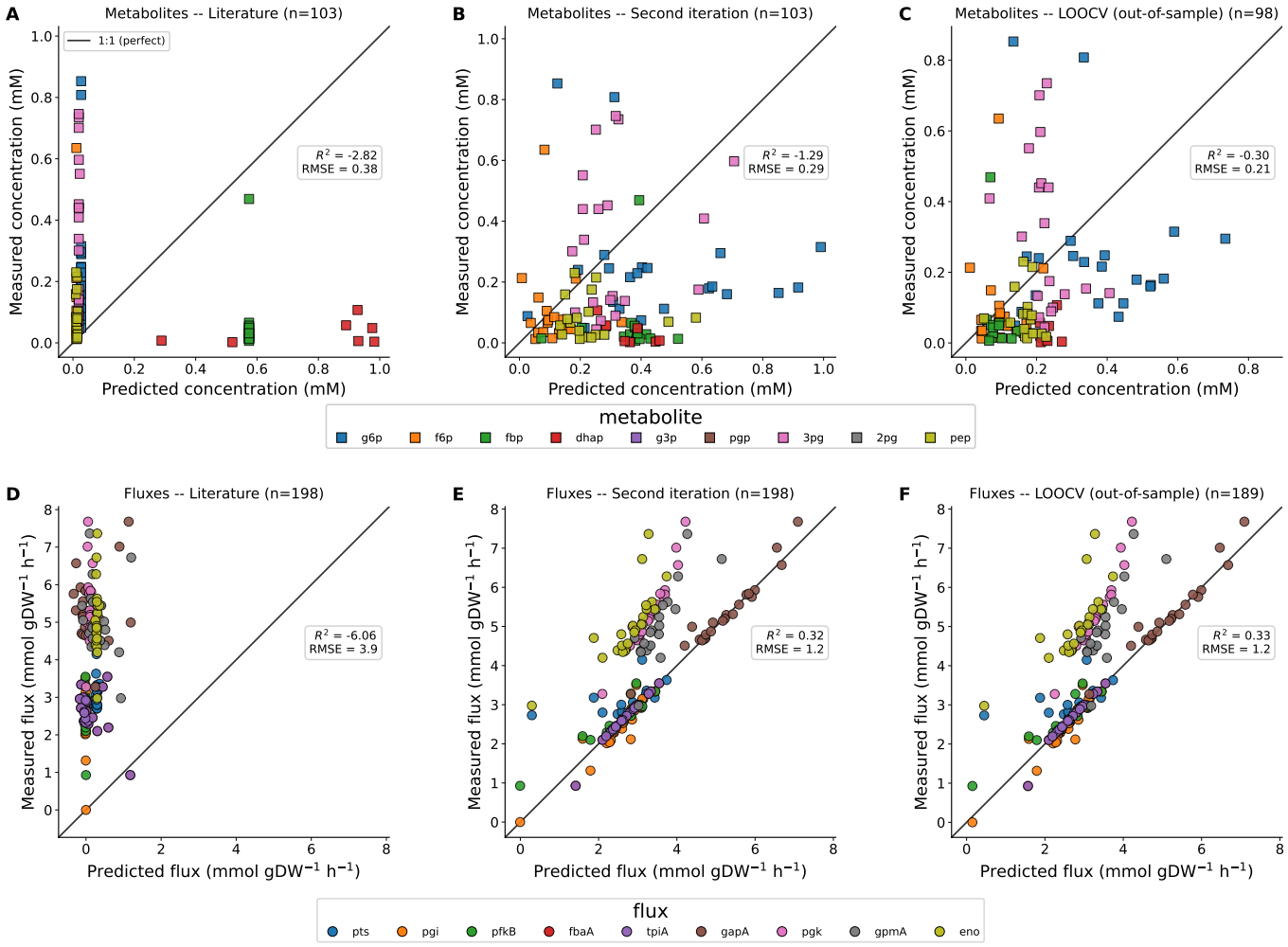


Figure S6: Comparison of model predictions across the reduction and validation pipeline. Parity plots of measured vs. predicted metabolite concentrations (**A–C**) and fluxes (**D–F**) using literature parameter values (**A, D**), the final reduced model fitted on the full training set (**B, E**), and leave-one-out cross-validation predictions on held-out conditions (**C, F**). Point colors indicate individual metabolites (**A–C**) or reactions (**D–F**).

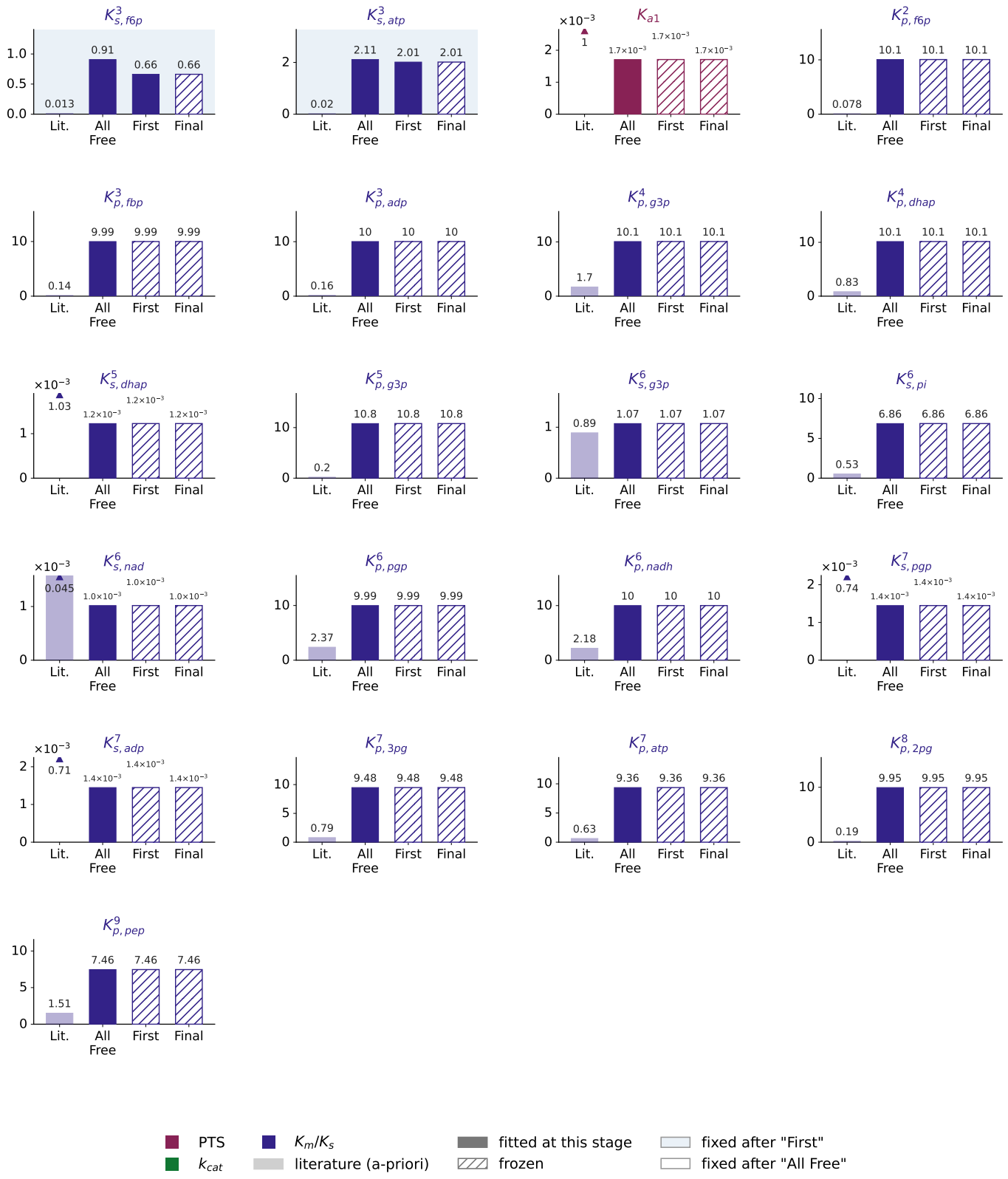


Figure S7: Evolution of parameter values across the four estimation stages: literature, initial free estimation, first reduced model, and final reduced model. Bar color indicates parameter type (PTS, k_{cat}, or K_m/K_s); fill pattern indicates the parameter's status at each stage: literature a-priori value, actively fitted, frozen at a previously fitted value, or newly fixed after the first or second reduction step.

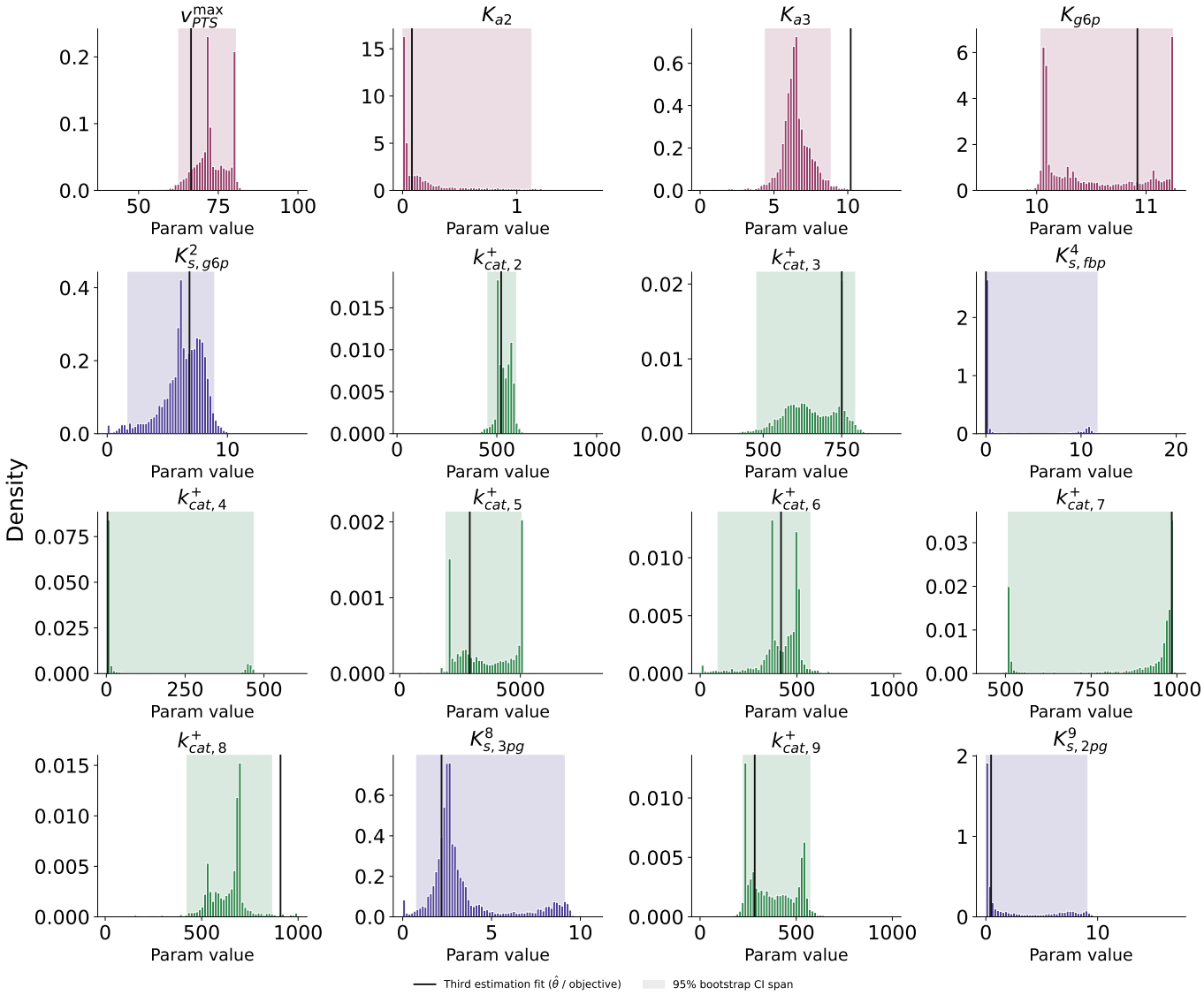


Figure S8: Bootstrap distributions ($n = 5000$) of the 16 free parameters of the final reduced model, obtained by resampling experimental conditions and re-solving the parameter estimation problem. The solid line marks the parameter estimate from the full dataset; the shaded region marks the 95% bootstrap confidence interval. Bimodal or heavy-tailed distributions indicate parameters poorly constrained by the available data.

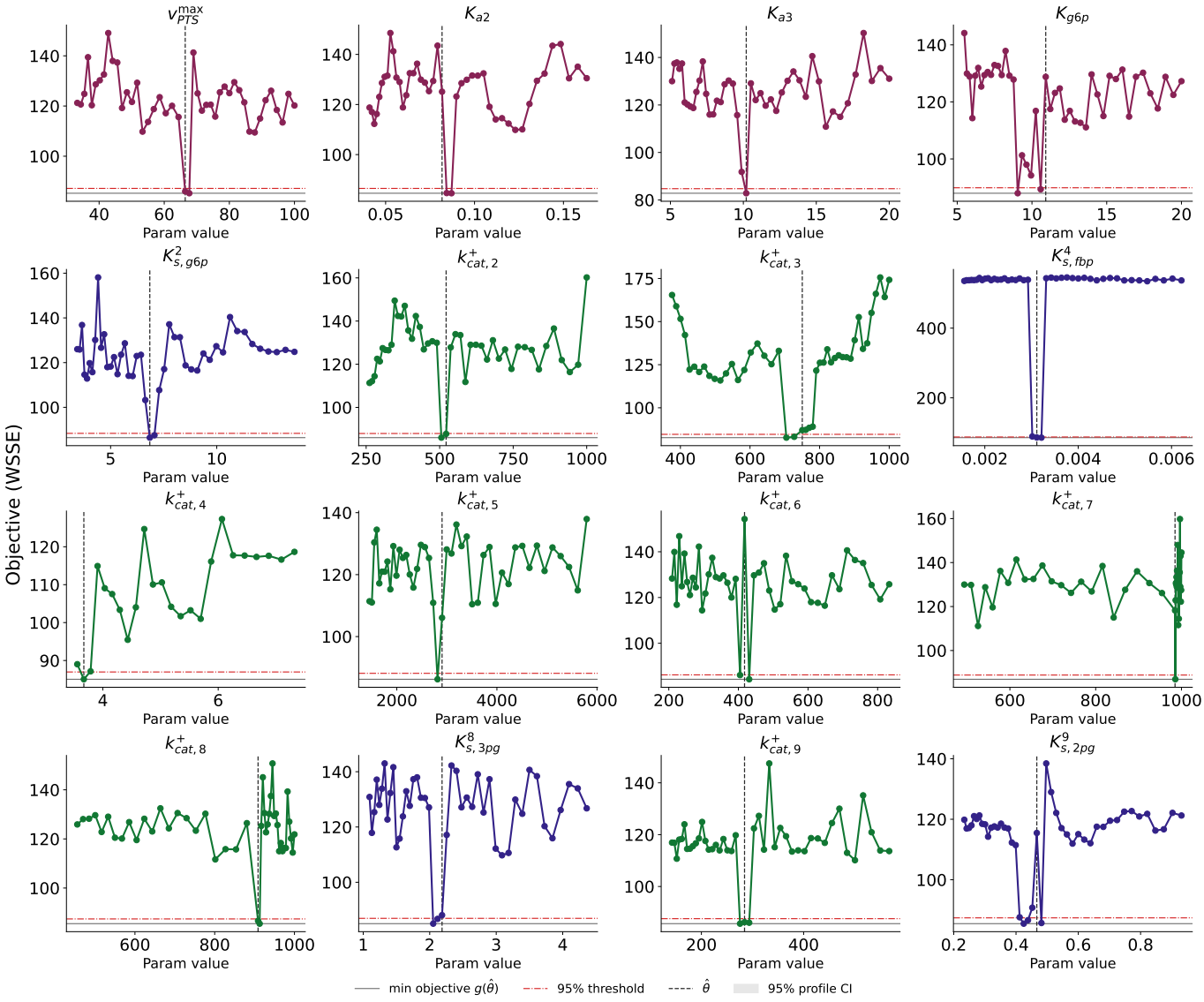


Figure S9: Profile likelihood of the 16 free parameters of the final reduced model. Each panel shows the objective function (WSSE) evaluated over a grid of fixed values for the corresponding parameter, with all remaining parameters re-optimized at each point. The dash-dotted line marks the 95% confidence threshold $\Delta(\alpha)$ above the global minimum; the dashed vertical line marks the estimated value $\hat{\theta}$. Panels with multiple points below the threshold reveal alternative local optima reached by the optimizer.

S-III. TABLES

Table I: Parameter descriptions

Symbol	Meaning
e_i	Enzyme concentration for reaction i
$k^+_{cat,i}$	Forward turnover number for reaction i
$k^r_{s,i}$	Michaelis-Menten constant for substrate i in reaction r.
$k^r_{p,i}$	Michaelis-Menten constant for product i in reaction r.
K^r_{eq}	Equilibrium constant for reaction r.
$[s]$	Substrate concentration
$[p]$	Product concentration.

Table II: Metabolites involved, abbreviation and their mass-balance classification

Abbrev.	Metabolite	Classification
glc	Glucose	Imbalanced
$g6p$	Glucose 6-phosphate	Balanced
$f6p$	Fructose 6-phosphate	Balanced
fbp	Fructose 1,6-bisphosphate	Balanced
$dhap$	Dihydroxyacetone phosphate	Balanced
$g3p$	Glyceraldehyde 3-phosphate	Balanced
pgp	1,3-Bisphosphoglycerate	Balanced
$3pg$	3-Phosphoglycerate	Balanced
$2pg$	2-Phosphoglycerate	Balanced
pep	Phosphoenolpyruvate	Balanced
pyr	Pyruvate	Imbalanced
atp	ATP	Imbalanced
adp	ADP	Imbalanced
nad	NAD^+	Imbalanced
$nadh$	NADH	Imbalanced
pi	Inorganic phosphate	Imbalanced

Table III: Glycolysis kinetic parameters in *E. coli* obtained from EcoCyc[10], Kadir et al. [12] and Equilibrator [11].

#	Enzime (gen)	Reaction	Parameters	Source
1	Phosphotransferase system	$glc + pep \rightarrow g6p + pyr$	v^{PTS}_{max}: 25.739 mmol/(gDCW·h) K_{a1}: 1.0 mM K_{a2}: 0.01 mM K_{a3}: 1.0 mM n_{g6p}: 4 K_{g6p}: 0.5 mM	[12]
2	Phosphoglucose isomerase (*Pgi*)	$g6p \rightleftharpoons f6p$	K_{eq} : 0.381 K^{g6p}_m: 1.018 mM K^{f6p}_m: 0.078 mM k_{cat} : $--$	EcoCyc: EG10702
3	Phosphofructokinase (*PfkB*)	$f6p + atp \rightarrow fbp + adp$	K_{eq} : 341.298 K^{f6p}_m: 0.013 mM K^{atp}_m: 0.02 mM K^{fbp}_m : 0.14 mM K^{adp}_m : $--$ k_{cat} : $--$	EcoCyc: EG10700

Continues on next page

Table III – continuation

#	Enzime (gen)	Reaction	Parameters	Source
4	Aldolase (*FbaA*)	$fbp \rightleftharpoons g3p + dhap$	K_{eq} : 0.1616 K_m^{fbp}: 0.24 mM K_m^{g3p} : $--$ K_m^{dhap} : $--$ k_{cat} : $10.33s^{-1} \rightarrow 0.95\frac{mmol}{mg \cdot h}$	EcoCyc: EG10282
5	TPI (*TpiA*)	$dhap \rightleftharpoons g3p$	K_{eq} : 0.113 K_m^{dhap}: 1.03 mM K_m^{g3p} : $--$ k_{cat}: 9000 $s^{-1} \rightarrow 1201.2\frac{mmol}{mg \cdot h}$	EcoCyc: EG11015
6	GAPDH (*GapA*)	$g3p + nad + pi \rightleftharpoons pgp + nadh$	K_{eq} : 0.00026 K_m^{g3p}: 0.89 mM K_m^{nad}: 0.045 mM K_m^{pi}: 0.53 mM K_m^{pgp} : $--$ K_m^{nadh} : $--$ k_{cat} : $--$	EcoCyc: EG10367
7	Phosphoglycerate kinase (*Pgk*)	$pgp + adp \rightleftharpoons 3pg + atp$	K_{eq} : 1849.572 K_m^{pgp} : $--$ K_m^{adp} : $--$ K_m^{3pg} : $--$ K_m^{atp} : $--$ k_{cat} : $--$	Equilibrator [11]
8	Phosphoglycerate mutase (*GpmA*)	$3pg \rightleftharpoons 2pg$	K_{eq} : 0.1874 K_m^{3pg}: 0.2 mM K_m^{2pg}: 0.19 mM k_{cat}: 330 $s^{-1} \rightarrow$ 41.602 $\frac{mmol}{mg \cdot h}$	EcoCyc: EG11699
9	Enolase (*Eno*)	$2pg \rightleftharpoons pep$	K_{eq} : 4.5273 K_m^{2PG}: 0.1 mM K_m^{pep} : $--$ k_{cat} : $--$	EcoCyc: EG10258

Table IV: Parameter values across estimation stages for the 37-parameter E. coli glycolysis model.

Parameter	Literature parameters	All Free	First reduced model	Final reduced model	Fixed/Free
$v_{PTS}^{\max}$	25.7	81	80.3	66.4	Free
K_{a1}	1	1.71×10^{-3}	1.71×10^{-3}	1.71×10^{-3}	Fixed
K_{a2}	0.01	0.041	0.0297	0.0817	Free
K_{a3}	1	5.3	6.47	10.2	Free
K_{g6p}	0.5	11.3	11.2	10.9	Free
$K_{s,g6p}^{2}$	1.02	6.97	6.18	6.85	Free
$K_{p,f6p}^{2}$	0.078	10.1	10.1	10.1	Fixed
$k_{cat,2}^{+}$	21.8	501	505	522	Free
$K_{s,f6p}^{3}$	0.013	0.908	0.662	0.662	Fixed
$K_{s,atp}^{3}$	0.02	2.11	2.01	2.01	Fixed
$K_{p,fbp}^{3}$	0.14	9.99	9.99	9.99	Fixed
$K_{p,adp}^{3}$	0.163	10	10	10	Fixed
$k_{cat,3}^{+}$	7.37	751	753	750	Free
$K_{s,fbp}^{4}$	0.24	1.03×10^{-3}	3.14×10^{-3}	3.11×10^{-3}	Free
$K_{p,g3p}^{4}$	1.7	10.1	10.1	10.1	Fixed
$K_{p,dhap}^{4}$	0.83	10.1	10.1	10.1	Fixed
$k_{cat,4}^{+}$	0.95	3.56	3.67	3.67	Free
$k_{cat,5}^{+}$	1.20×10^{3}	916	2.05×10^{3}	2.90×10^{3}	Free
$K_{s,dhap}^{5}$	1.03	1.23×10^{-3}	1.23×10^{-3}	1.23×10^{-3}	Fixed
$K_{p,g3p}^{5}$	0.197	10.8	10.8	10.8	Fixed
$k_{cat,6}^{+}$	77.6	348	368	418	Free
$K_{s,g3p}^{6}$	0.89	1.07	1.07	1.07	Fixed
$K_{s,pi}^{6}$	0.53	6.86	6.86	6.86	Fixed
$K_{s,nad}^{6}$	0.045	1.02×10^{-3}	1.02×10^{-3}	1.02×10^{-3}	Fixed
$K_{p,pgp}^{6}$	2.37	9.99	9.99	9.99	Fixed
$K_{p,nadh}^{6}$	2.18	10	10	10	Fixed
$k_{cat,7}^{+}$	23.9	983	984	985	Free
$K_{s,pgp}^{7}$	0.738	1.45×10^{-3}	1.45×10^{-3}	1.45×10^{-3}	Fixed
$K_{s,adp}^{7}$	0.71	1.45×10^{-3}	1.45×10^{-3}	1.45×10^{-3}	Fixed
$K_{p,3pg}^{7}$	0.792	9.48	9.48	9.48	Fixed
$K_{p,atp}^{7}$	0.632	9.36	9.36	9.36	Fixed
$k_{cat,8}^{+}$	41.6	562	694	909	Free
$K_{s,3pg}^{8}$	0.2	3.6	2.62	2.19	Free
$K_{p,2pg}^{8}$	0.19	9.95	9.95	9.95	Fixed
$k_{cat,9}^{+}$	48.8	248	237	284	Free
$K_{s,2pg}^{9}$	0.1	0.453	0.233	0.467	Free
$K_{p,pep}^{9}$	1.51	7.46	7.46	7.46	Fixed